

Roll No.

TBT-501

**B. TECH. BIOTECHNOLOGY/B. TECH. BIOTECHNOLOGY
(HONS.) WITH SPECIALIZATION IN COMPUTER SCIENCE
AND BIOLOGY (FIFTH SEMESTER)**

END SEMESTER EXAMINATION, Dec., 2022

RECOMBINANT DNA TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Biosafety ? Describe characteristics of BSL3 with a layout. (CO1)
(b) What is gene cloning ? Write down its applications in biological sciences. (CO1)
(c) Write short notes on the following : (CO1)
 - (i) Scope of Genetic Engineering
 - (ii) History of RDT
2. (a) What are restriction nucleases ? Why are only Type-II restriction endonucleases preferred in ROT ? (CO2)

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- (b) How modifying enzymes are helpful in gene cloning ? (CO2)
- (c) Write short notes on the following : (CO2)
- (i) Ti vector- pBIN19
 - (ii) Phagemid
3. (a) Explain cloning and screening strategies in COSMID vector- pJB8. (CO3)
- (b) Describe principal and application of Red-White Screening in YAC vectors. (CO3)
- (c) With the help of suitable diagram, describe any pET vector along with mechanism of regulation of its expression. (CO3)
4. (a) What are DNA markers ? Describe any *two* with their applications. (CO4)
- (b) What is the principle of PCR ? Write down different variants of PCR. (CO4)
- (c) Write short notes on the following : (CO4)
- (i) Sanger's method of DNA sequencing
 - (ii) DNA probes
5. (a) What is DNA barcoding ? How is it useful in phylogenetic relationships ? (CO1)
- (b) Describe principle and applications of CRISPR-CAS technology. (CO1)
- (c) Write short notes on the following : (CO1)
- (i) Transgenic plants
 - (ii) Gene therapy

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B. TECH. (BIOTECHNOLOGY)

(FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

HEAT AND MASS TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Biot Number ? How is it different from Nusselt number ? Give a reasonable statement. What happen to the Biot number if the conductance resistance increases to infinity ? (CO1/CO2)
- (b) What is the critical radius of insulation for asbestos with $K = 0.17 \text{ W/m-K}$ surrounding a circular pipe and exposed to room air at 20°K with heat transfer coefficient of $3 \text{ W/m}^2\text{-K}$. (CO1)

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- (c) What will be the critical radius for asbestos with $K = 0.125 \text{ W/m-K}$ surrounding a pipe exposed to room air with $h = 2.5 \text{ W/m}^2\text{-K}$. (CO1)
2. (a) Find the heat transfer through infinite long fin with the following data :
Heat transfer coefficient (h) = $40 \text{ W/m}^2\text{-K}$, Perimeter (P) = 8 m , thermal conductivity (K) = 3.5 W/m-K , cross section area (A) = 4 m^2 . Base temperature ($T_o = 200^\circ\text{K}$) and exposed to environment at 50°K . (CO2)
- (b) Define fin efficiency. How it varies with respect to base temperature ? Calculate the fin efficiency having actual heat transfer of 350 W and maximum possible heat transfer of 500 W . (CO2)
- (c) Define fin effectiveness. What would happen if number of fins are increased in the system ? Calculate the fin effectiveness if heat transfer with fin is 88 W and heat transfer without fin is 48 W . (CO2)
3. (a) Explain the physical significance of Reynolds number, Stanton number and Prandtl number. (CO1/CO3)
- (b) Find the Reynolds number of the flowing fluid inside a pipe with density = 120 kg/m^3 , velocity = 30 m/sec , diameter of pipe = 2 m and viscosity = 4.5 N-s/m^2 . (CO3)
- (c) What is hydrodynamic and thermal boundary layer ? What is the relation between them ? Explain with the help of diagram. (CO3)
4. (a) Two large plates are maintained at a temperature of 900 K and 500 K respectively. Each plate has area of 62 . Compare the net heat exchange between the plates for the following cases : (CO4)
- (i) Both plates are black
- (ii) Plates have an emissivity of 0.5 .

- (b) The sun emits maximum radiation at $\lambda = 0.52 \mu$. Assuming the sun to be blackbody, calculate the surface temperature of the sun. Also calculate thermonochromatic emissive power of the sun's surface : (CO4)

Given :

$$\lambda_{\max} = 0.52, \mu = 0.52 \times 10^{-6} \text{ m}$$

To find :

- (i) Surface temperature T
 - (ii) Monochromatic emissive power, $E_{b\lambda}$
 - (iii) Total emissive power, E
 - (iv) Maximum emissive power, $E_{\max}$
- (c) Define the following with equation : (CO4)
- (i) Absorptivity
 - (ii) Reflectivity
 - (iii) Transmittivity
 - (iv) Irradiation

5. (a) Write down and explain the NTU in parallel flow and counter flow of heat exchanger. Define the concept of minimum specific heat used in NTU method. (CO5)

- (b) Water flows at the rate of 65 kg/min through a double pipe counter flow heat exchanger. Water is heated from 50°C to 75°C by an oil flowing through the tube. The specific heat of the oil is 1.780 kJ/kgK. The oil

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enters at 115°C and leaves at 70°C . The overall heat transfer coefficient is $340 \text{ W/m}^2\text{K}$. Calculate the following : (CO5)

- (i) Heat exchanger area
- (ii) Rate of heat transfer
- (c) What is latent heat ? How is it different from sensible heat ? How can you calculate the losses occur in heat exchanger ? Explain briefly.

(CO5)

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**B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

BIOPROCESS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the historical development and progression of Fermentation Industry. (CO1)
- (b) Describe the steps of media optimization as per Plackett Burman design along with example. (CO1)
- (c) What is Del factor ? Which equation relates Del with Time and Temperature of Sterilization ? Derive that equation. (CO1)
2. (a) The growth of *S.cerevisae* on glucose results in production of ethanol. Calculate the degree of reduction of substrate glucose and ethanol. What is Leudking piret equation and Monod equation ? (CO2)
- (b) Derive an expression for total Biomass, substrate and Product concentration in continuously stirred tank reactors. (CO2)
- (c) What is Double Reciprocal plot ? Explain product profile in Batch culture using suitable equations and graphs. (CO2)

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3. (a) Describe Oxygen Balance Technique and Dynamic gassing out method used for estimation of K_La in bioreactors. (CO3)
- (b) How do Temperature, Pressure, Power input to agitator, Gas velocity and Bubble size affect oxygen transfer from gas bubble to cell site in fermenter ? (CO3)
- (c) *E.coli* is used for production of proline amino acid. Q_{O_2} of *E.coli* in batch culture is $5 \text{ mmol O}_2 \text{ g}^{-1} \text{ h}^{-1}$. The bacteria is grown in a stirred fermenter to a cell density of 40 g/l ; K_{La} is 0.15 per second. At the fermenter operating temperature and pressure, the solubility of oxygen in the culture liquid is $8 \times 10^{-3} \text{ kg/m}^3$. Is the cell metabolism limited by mass transfer, or dependent solely on metabolic kinetics ? (CO3)
4. (a) Differentiate between Solid state and Submerged Fermentation. (CO4)
- (b) Describe fermentative production of Penicillin. Explain applications of penicillin. (CO4)
- (c) Discuss fermentative production of amylases using agro-industrial wastes. (CO4)
5. (a) Discuss role of Biosensors and Artificial neural network in Bioprocess Industry. (CO5)
- (b) Describe different methods of immobilization of cells. What is the advantage of immobilization of cells and enzymes ? (CO5)
- (c) How do you preserve industrially important microbes ? What is application of lipases, proteases, cellulases and pectinases in industry ? (CO5)

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**B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

BIOINFORMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define bioinformatics in your own words. How can bioinformatics be used for sequence analysis ? (CO1)
- (b) Describe major hardware components required for proper functioning of computer along with diagram. (CO1)
- (c) Describe major software components required for proper functioning of computer along with diagram. (CO1)
2. (a) Throw light on any two techniques that generate data for bioinformatics. (CO1, CO2, CO5)
- (b) What are the various criteria for classification of databases ? Also give example from each category. (CO1, CO2, CO5)

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- (c) How does BLAST algorithm work ? What are its various applications ?
(CO1, CO2, CO5)
3. (a) Describe homology modeling as a tool for protein structure prediction.
(CO3, CO4, CO5)
- (b) What are the various secondary RNA structures ? Explain with proper diagram.
(CO3, CO4, CO5)
- (c) Write short notes on the following : (CO3, CO4, CO5)
- (i) Drug designing
- (ii) Rooted and Unrooted phylogenetic trees
4. (a) Differentiate between local and global sequence alignment along with proper diagram. (CO4, CO5)
- (b) Elaborate the role of network biology in understanding human diseases.
(CO4, CO5)
- (c) Throw light on any genome editing tool studied by you and also how computational tools can assist in the same. (CO4, CO5)
5. (a) What do you understand by gene therapy ? How can bioinformatics help in gene therapy ? (CO4, CO5)
- (b) Throw light on GWAS and its application in understanding human diseases ? (CO4, CO5)
- (c) Elaborate the concept of personalized medicines and role of bioinformatics/computational biology in understanding it. (CO4, CO5)

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**B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

STEM CELL TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the key concept involved for the phenotypic changes that do not involve alterations in the DNA sequence. (CO3 & CO4)
- (b) Highlight the role of intermediate metabolites in stem cell technology. (CO3 & CO4)
- (c) Discuss the importance of genetically engineered stem cells. (CO3 & CO4)
2. (a) What is oxidative stress ? How the oxidative stress is involved in pathological process and physiological process. (CO3 & CO4)
- (b) Discuss the steps involved in the culturing of embryo. Also explain the Ethical issues pertaining to Stem cell therapy. (CO3 & CO4)
- (c) How do epigenetic mechanisms regulate gene expression ? (CO3 & CO4)

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3. (a) "The process of fertilization where an egg is combined with sperm in vitro". Explain the technique involved and explain the molecular events followed during the aforementioned procedure. (CO4 & CO1)
- (b) How the long term storage of the stem cells could be carried out ? Explain the procedure followed for the same with examples. (CO4 & CO1)
- (c) Highlight the relevance of fate mapping in embryonic development. (CO4 & CO1)
4. (a) Explain the tests used for the identification of stem cells with diagram. (CO2 & CO5)
- (b) How stem cell could be used as a weapon for an efficient drug delivery Process ? (CO2 & CO5)
- (c) Describe the techniques for the identification of stem cells. (CO2 & CO5)
5. (a) "The process that aids in the development of neuron." Explain the process with markers associated with it. (CO1 & CO2)
- (b) Differentiate between tissue engineering and genetic engineering. (CO1 & CO2)
- (c) How Mesenchymal stem cells aid in the regeneration process ? (CO1 & CO2)

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TBT-504(c)

B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

GENOME EDITING TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the repair pathway which often gets activated during DNA double strand break. (CO1)
- (b) Write a note on : cre-loxP and Flp-FRT system of recombination. (CO1)
- (c) Elucidate the process and steps involved in miRNA based RNA silencing. (CO1)
2. (a) Define : ZFNs, TALENs, Meganucleases. (CO2)

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- (b) What is CRISPR ? Describe its all components and their role in gene editing ? (CO2)
- (c) CRISPR does not harm the genome of bacterial cells. Elaborate. (CO2)
- 3. (a) What is MAGE ? Why it is required and what is the makeup of the machine ? (CO3)
- (b) What is SgRNA ? Describe the parameters need to be considered while designing SgRNA ? (CO3)
- (c) Describe any *three* software of SgRNA designing ? (CO3)
- 4. (a) How gene editing techniques is used for creating transgenic plants ? (CO4)
- (b) Describe the need for endogenous gene labelling and chromosomal rearrangement. (CO4)
- (c) Write about three application of genome editing techniques for bioremediation. (CO4)
- 5. (a) Describe the aim, scope and objectives of National Guidelines for Gene Therapy Product Development and Clinical Trials, 2019. (CO5)
- (b) What are the focus area of National Guidelines for Stem Cell Research, 2017 ? Who gave these guidelines and why ? (CO5)
- (c) Discuss the ethical issues associated with the application of CRISPR on human cells. (CO5)

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B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

INTERNET OF THINGS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Flutura employs artificial intelligence to improve asset uptime and operation efficiency. Using its Cerebra Industrial AI and IoT platform, the company serves clients in the manufacturing, oil and gas and utilities industries. Design an IoT-based architecture for Flutura. (CO1)
- (b) ScienceSoft provides comprehensive IoT services for 30+ industries since 2011. They design and build robust IoT solutions for production control and optimization, asset and employee tracking, remote health monitoring, transportation management as well as smart connected products, smart homes, and smart cities. Discuss the role of IoT in the perspective of ScienceSoft. (CO1)

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- (c) iTechArt Group is a custom software development company with a heavy focus on delivering feature-rich and secure IoT solutions for startups and fast-growing tech companies. From GPS tracking to building smart secure home solutions, the iTechArt team steps in as a goto partner for businesses willing to build fully integrated IoT solutions. Classify the sensors used for iTechArt Group. (CO1)
2. (a) Health monitoring and remote diagnosis can be realized through Smart Healthcare. List the ten Smart healthcare applications. Analyse the applications of IoT in Smart healthcare sector. (CO1)
- (b) Oxagile is a professional IoT company that offers a full range of services including IoT consulting, Software development, Hardware prototyping, Integration, and Continuous enhancement, Apply the key technologies of Internet of Things in the perspective of Oxagile. (CO2)
- (c) SumatoSoft has been building IoT solutions since 2012, helping enterprises and startups gain a competitive advantage, and improve their efficiency, effectiveness, and prot through business digitalization. Illustrate the sensor technology in the perspective of SumatoSoft. (CO2)
3. (a) Andersen provides custom IoT development services for businesses of all sizes and specializes in helping companies develop and deploy custom IoT solutions that improve efficiency, communication, and data collection, Sketch the WSN architecture in the perspective of Andersen's IoT platform. Also, describe the role of WSN in Andersen perspective. (CO3)

- (b) Impinj specializes in solutions involving radio-frequency identification, or RFID. Impinj's RFID tags are used by retailers, manufacturers, and logistics companies to track inventory and assets. Use the components of RFID in the perspective of Impinj. (CO3)
- (c) Dexcom is focused on diabetes management, specifically medical devices for continuous glucose monitoring. The company's G6 system includes an auto a sensor and transmitter, and a touchscreen receiver displaying real-time glucose data. Use the RFID technologies in Smart health applications of IoT. (CO3)
4. (a) Wipro's Internet of Things solutions addresses the connectivity of 'legacy' & 'new' things. Our Multi-Protocol IoT Gateway Framework blends heterogeneous protocols that complement IoT at the edge, enabling sele data to be transmitted to backend systems. Interpret the data transmission and security challenges in the perspective of Wipro. (CO4)
- (b) Algo Engines is a leading internet of things company working in Industrial IoT (IIoT) in India. It works on building actionable insights by analyzing the data provided by IoT. Algo engines is a company that works on renewable energy assets like met masts, wind turbines, wind farms, solar plants, and hydro projects. Identity the management model of Algo Engines. (CO5)
- (c) Alitux Innovations is a leading internet of things company in India. It is a software and product engineering services organization that focuses on developing Internet of Things (IoT) products and services for multiple domains like smart cities, smart homes, industrial IoT, intelligent buildings, semiconductors and many more. Apply the resource management in the internet of things with the perspective of Alitux Innovations. (CO5)

5. (a) Altizon Systems is a Wipro-backed internet of things company in India which develops solutions based on IoT. Altizon has over 100 Enterprise users, making it a Leading Industrial IoT platform provider. Recommend the use of Software Agent in the perspective of Altizon Systems. (CO5)
- (b) Clover Health is a Preferred Provider Organization (PPO) and a Health Maintenance Organization (HMO) with a Medicare contract. Create a Body Area Network (BAN) for Clover Health's user and identify the components of BAN in the perspective of Clover Health. (CO6)
- (c) Nexleaf provides data analytics tools and lightweight sensor services to those in the global public health sector. A nonprofit company, its team of computer scientists, software engineers, public health experts and many others works extensively in low-and middle-income countries. Identify the security requirements of IoT applications in the perspective of Nexleaf. (CO6)

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TBT-506(b)

B. TECH. BIOTECHNOLOGY (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

NETWORK BIOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the main principles of Graph theory ? (CO1, CO2)
(b) Describe any two networks in Biology along with its applications. (CO1, CO2)
(c) What are the various kinds of area networks in computer networks ?
Elaborate with proper diagram. (CO1, CO2)
2. (a) Elaborate the terms : (CO1, CO2, CO5)
 - (i) Lethality in protein networks
 - (ii) Degree of network
 - (iii) Herpes viral protein network
 - (iv) Scale free Networks
 - (v) Centrality
(b) Throw light on STRING and STICH FOR understanding molecular interactions. (CO1, CO2, CO5)

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- (c) What is meant by protein-protein interaction and the need for graphical representation of protein networks ? (CO1, CO2, CO5)
3. (a) What are the various classes of metabolic reactions in the organism ? Give example from each category. (CO3, CO4, CO5)
- (b) Throw light on the statement 'The small world inside large metabolic networks'. (CO3, CO4, CO5)
- (c) Write short notes on the following : (CO3, CO4, CO5)
- (i) Databases for understanding metabolic networks
- (ii) Metabolic fluxes in *E. coli*
4. (a) What are gene co-expression networks ? How are they different from gene regulatory networks ? (CO3, CO4, CO5)
- (b) Elaborate transcriptional regulatory circuits in human embryonic stem cells. (CO3, CO4, CO5)
- (c) With the help of diagram explain structure of a gene regulatory network (GRN). Also throw light on working of GRN. (CO3, CO4, CO5)
5. (a) What are ecological networks ? Write any *four* characteristics of ecological networks. (CO4, CO5)
- (b) Throw light on drug interaction networks. What are the applications of drug interaction networks ? (CO4, CO5)
- (c) With the help of well labeled diagrams describe food web and its importance in maintaining life. (CO4, CO5)

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TBT-505(B)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

3D PRINTING AND DESIGNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain Fused Deposition Modelling in detail with a suitable example.

(CO1)

(b) Explain how additive manufacturing changes current scenario of manufacturing world.

(CO1)

(c) Classify additive manufacturing systems. Explain each system in detail with proper diagram and labeling.

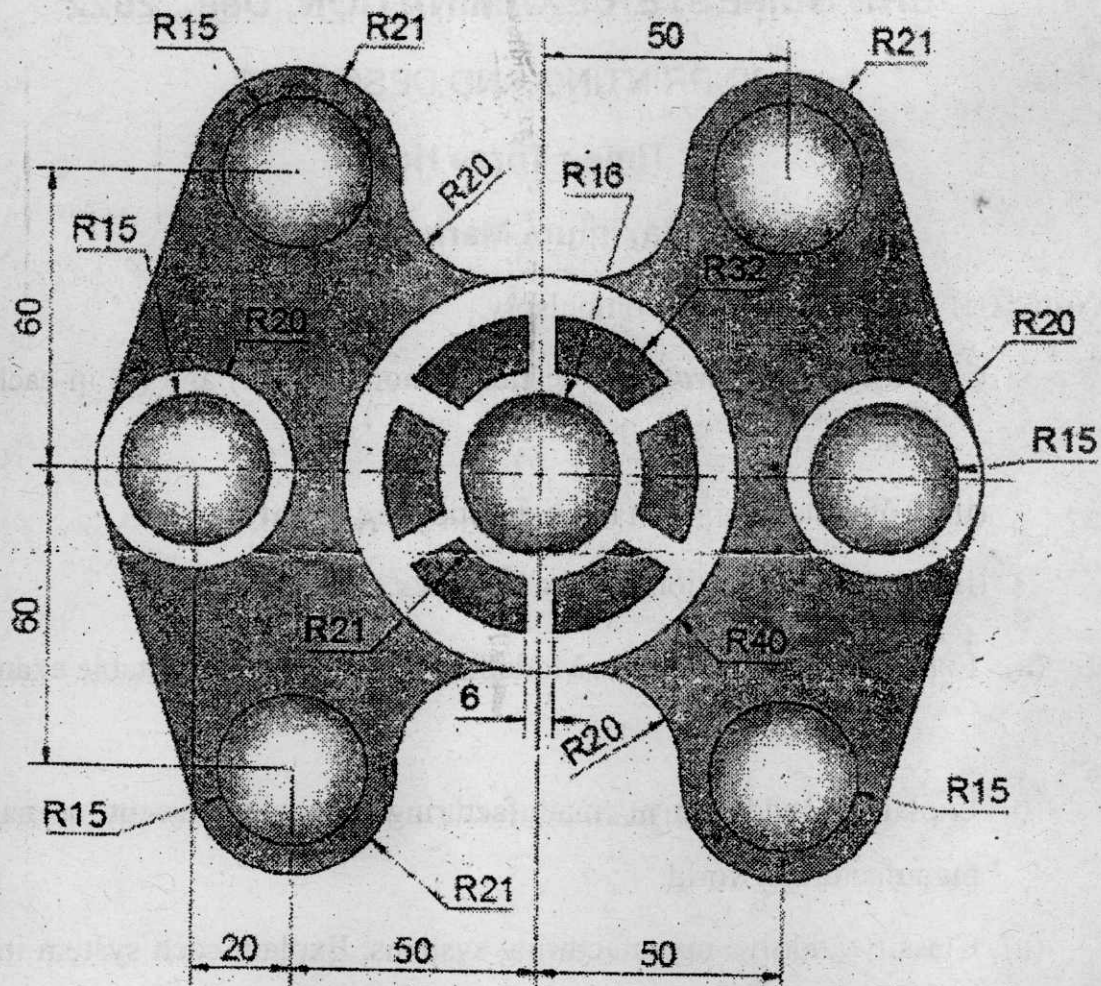
(CO1)

2. (a) What you mean by CAD ? Explain how CAD is helpful in manufacturing as well as in reverse engineering.

(CO2)

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- (b) In which fields do you see the maximum use of designing software ?
Explain with suitable example. (CO2)
- (c) Write the steps involved in making given drawing on AutoCad software : (CO2)



3. (a) What is A 3d Printing Model ? What are the most common 3d Printing file formats ? (CO3)

(3)

- (b) How can I reduce my 3d printing costs ? Which materials can't be 3d printed ? (CO3)
- (c) What is not suitable for 3d printing ? Is 3d printing food possible ? (CO3)
- 4. (a) How is SLS method of 3D printing different from FDM ? (CO4)
- (b) What is the limitation in SLS process ? What material is used for SLS printing ? (CO4)
- (c) What are applications of SLS type 3D printers ? Which type of 3D printing technique is mostly use in medicine field ? (CO4)
- 5. (a) Write the parameters for making a 3D printed model from using SLS technique with proper diagram. (CO5)
- (b) Write the parameters for making a 3D printed model from ABS filament with diameter of 1.75 mm by using FDM technique. (CO5)
- (c) Discuss with flowchart to make a project from beginning to the final 3D printed model. (CO5)